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The Khao Yai Fault on the southern margin of the Khorat Plateau, and the pattern of faulting in Southeast Thailand

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ABSTRACT

A study of Google Earth images has revealed a hitherto-unrecorded gently curved lineament within the southern marginal zone of the Khorat Plateau in eastern Thailand. The lineament, confirmed by digital elevation model (DEM) images, is at least 130 km long and coincides with a dip reversal of the Mesozoic Khorat Group. It is interpreted here as a fault, named the Khao Yai Fault, and it has characteristics which make it unusual within the Khorat Plateau. The fault forms the northern boundary of a belt of several ENE–WSW trending fault splays which are thought to link with the Mae Ping Fault further south; this is interpreted as a left-stepping, sinistral strike-slip duplex about 50 km wide and 150 km long. Apatite fission track data indicate that exhumation began during the earliest Palaeogene.

The Khao Yai Fault is considered in its regional context which includes the Cardamomes Mountains of Cambodia, the offshore Phuquoc-Kampot Basin, and the Khao Thalai Red-beds outlier of the Khorat Group in Southeast Thailand. The latter is interpreted as a down-faulted sliver of the Khorat Group in the Tha Mai Fault belt which is thought, in turn, to be a splay of Thailand's other major regional fault, the Three Pagodas Fault. Carboniferous, Permian and Triassic shallow-marine rocks with unusual faunas occur in a limited NNW–SSE trending zone to the west and NNW of the Tha Mai Fault and it is suggested that wrench movement on the fault played a part in the emplacement of these rocks.

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1. Introduction

Nearly 40 years ago it was shown that a study of topographic maps could be a powerful tool for recognizing major faults in Thailand. The result of that study (Ridd, 1971) was the discovery of the Three Pagodas and Mae Ping faults which are now viewed as key features of the structure of Southeast Asia. Satellite imagery is now routinely used to identify increasingly subtle structural features (e.g. Smith et al., 2007). But topographic maps still have a part to play and the present study was initially based on Google Earth images and more particularly Google terrain maps. Subsequently, other satellite images and digital elevation models (DEMs) were employed and interpreted on Global Mapper, supported by field observations. Global Mapper is a relatively simple, low-budget, easy-to-use geographic information system software package originally developed by USGS.[1] Google Earth has the advantage of ease of use, the ability to view anywhere in the world, and in places it has very high resolution. Its main disadvantages are the commonly contrasting patchwork of different satellite images, the distraction of vegetation cover, roads and buildings, and the lack of tools for mapping geological features which are available with DEM images, for example using different filters and display pallets. Interpreting images in Global Mapper, while zooming in to look at details of the geology on Google Earth in 3D view is a simple, but very effective way of combining the best of the two applications. Using this combination to map structural features in eastern Thailand, the presence of a hitherto unrecognized fault at least 130 km long within the southern border region of the Khorat Plateau has been revealed. It is called here the Khao Yai Fault as its fault scarp forms the northern boundary of the National Park of that name.

The Khorat Plateau occupies much of Northeast Thailand and is formed of Mesozoic and, locally, Palaeogene continental deposits with a total maximum thickness of at least 4000 m (Fig. 1). Much has been written on the rocks which form this 400 km wide plateau (e.g. Meesook, 2011; Meesook and Saengsrichan, 2011; Booth and Sattayarak, 2011; Buffetaut et al., 2009). The name Khorat Group has been variously applied to different groupings of formations in this non-marine succession (Fig. 2), for example the base of the Nam Pong Formation was adopted by Ward and

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[1] A free version with limited features is available as the USGS dlgv32 (http://mcmcweb.er.usgs.gov/drc/dlgv32pro). Free downloads of 1990/2000 coverage of Landsat data are available from NASA (https://zulu.ssc.nasa.gov/mrsid/), and 90m SRTM digital elevation data (DEMs) from the CGIAR consortium (http://srtm.csi.cgiar.org/).

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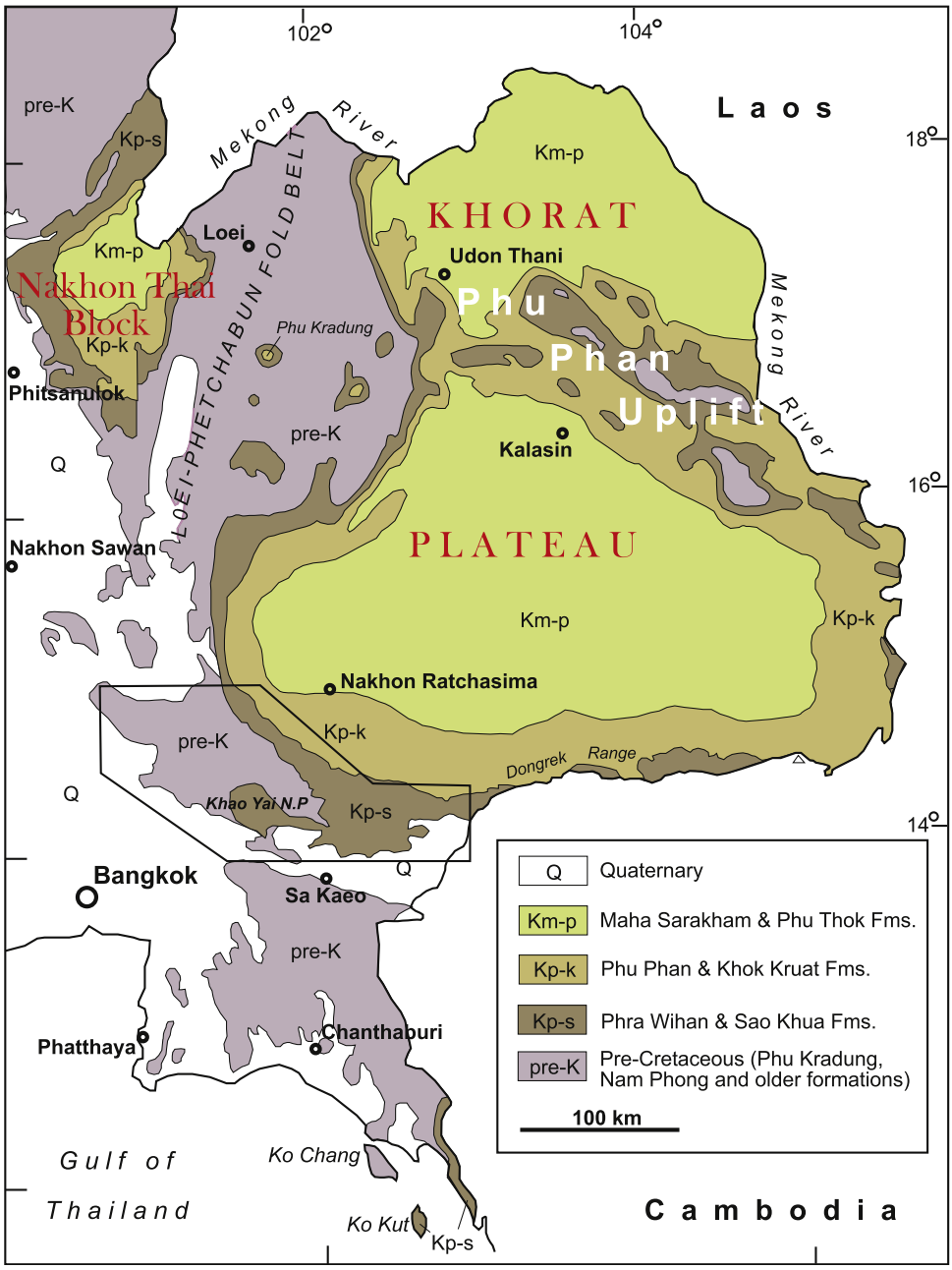


Fig. 1. Eastern Thailand. Simplified geological map based on the Department of Mineral Resources (1999), showing the extent of the Khorat Plateau, the type area of the variously defined Khorat Group. For simplicity, the Khorat Group outcrop of the Khao Thalai Ridge as well as some other small and less-reliably dated outcrop areas of Khorat Group in Southeast Thailand have been omitted. The outlined area NE of Bangkok is the area covered by Fig. 3a and b. NP = National Park.

Bunnag (1964) as the base of the group, whereas the base of the Kuchinarai Group (also called the Huai Hin Lat Formation) is the level adopted by Thailand's Department of Mineral Resources (DMR, 1999). But with petroleum exploration came the discovery that the succession includes several unconformities and this prompted, for example, Racey et al. (1994), Racey and Goodall (2009), and Booth and Sattayarak (2011) to restrict the term Khorat Group to the succession above the base of the Rhaetian Upper Nam Pong Formation (i.e. above the Indosinian II unconformity). Arguably there is a case for including the whole succession of non-marine beds from the base of the Kuchinarai Group to the top of the Phu Thok Formation as a single supergroup, notwithstanding the various unconformities that are present within and at the base of it (Fig. 2). Meanwhile, for convenience the term Khorat Group is adopted here in the sense used by the DMR, i.e. for the entire non-marine succession above the Indosinian I unconformity.

Over wide areas the rocks of the Khorat Plateau are flat-lying, but toward the plateau's western and southern rims the rocks dip inwards and are responsible for its broad dish-like structure. Also, crossing the northern part of the Khorat Plateau is a WNW–ESE trending belt of gentle anticlinal folding, the Phu Phan Uplift (Fig. 1). Petroleum exploration on the Khorat Plateau has shown that commonly the folds are accompanied at depth by faults within the lowest units of the Khorat Group but they die out upwards (e.g. Booth and Sattayarak, 2011). The Khao Yai Fault is thus unusual; here the younger units of the Khorat Group (the Phu Kradung and overlying Phra Wihan Formations) are affected.

Across the Khorat Plateau thickness changes are gradual, and even approaching the western margin of the plateau the Khorat

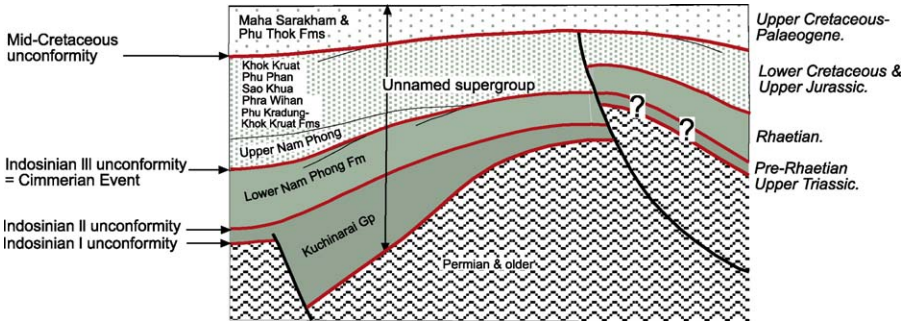


Fig. 2. Schematic stratigraphic relations of the non-marine succession in the area of the Khorat Plateau. The name Khorat Group has been applied variously to different groupings of units, but it is suggested here that a case exists for including the whole succession above the Indosinian I unconformity (the unconformity with the largest angularity) as a single supergroup; it is this succession to which the DMR applies the name Khorat Group. Indosinian unconformities I, II and III have been described by Booth and Sattayarak (2011) and Racey (2009). The mid-Cretaceous unconformity (Lovatt Smith et al., 1996) reflects a local episode of folding and faulting, but the question-mark symbols signify that the relations of the deeper units beneath such mid-Cretaceous faults are not known.

Group is still 2–3 km thick. Consequently the occurrence of only small outliers of Khorat Group west of the Plateau attests to a significant deformation event that caused widespread uplift and erosion of several kilometres of Khorat Group. This deformation is thought to be predominantly of Palaeogene age, based on apatite fission track dating of the exhumation (e.g. Upton, 1999; Racey et al., 1997; Morley, 2004; Morley et al., 2011). However, the style of deformation that caused the exhumation has not been described in detail, although Morley (2004) and Morley et al. (2007) suggest that a phase of transpressional deformation was responsible.

To investigate the regional setting of the Khao Yai Fault, the study has been widened to include the Cardamomes Mountains of southern Cambodia, the results of seismic reflection surveys in the Gulf of Thailand reported by Fyhn et al. (2010a), as well as integrating field observations in Southeast Thailand. The Khao Yai Fault and, in Southeast Thailand, the Tha Mai Fault help to validate the pattern of geophysical and other mapped lineaments which link these faults with, respectively, the major Mae Ping and Three Pagodas Fault belts.

The Mae Ping Fault belt trends NW–SE where it enters Thailand on the Myanmar border and then it swings to a N–S trend where it partly disappears beneath the Quaternary sediments of Thailand's central plain. Morley (2002, 2004) has proposed that the N–S trending hills projecting through the Quaternary plain are part of a strike-slip duplex system, and he suggested that the Mae Ping Fault then swings back to a WNW–ESE trend and continues into Cambodia where it underlies the Tonle Sap depression. Morley (2002, 2004), Morley et al. (2007), Smith et al. (2007) and Sangsomphong et al. (2008) assembled the evidence for this continuation based on the regional strike of scattered inliers in the broad Kabin Buri alluvial plain, gravimetric and aeromagnetic lineaments and apatite fission-track data. Importantly, the Khao Yai Fault provides the first direct evidence of Cretaceous or younger faulting on that WNW–ESE trend.

2. The Khao Yai Fault

Fig. 3a is a photogeological map produced from Google Earth terrain maps. Over much of the northern part of the map the Khorat Group dips northwards into the Khorat Basin, with prominent arcuate south-facing escarpments and northward dip-slopes. Equally conspicuous is the dip reversal running approximately WNW–ESE, with dip-slopes dipping SW and south, and north-facing escarpments aligned in a convex-southward line from about 101°35′E to 102°35E, a distance of about 130 km. The westernmost of these escarpments marks the northern boundary of Khao Yai National Park. Immediately north of these aligned north-facing escarpments is a topographically low-lying lineament which is interpreted here as a fault, named the Khao Yai Fault. South of the Khao Yai Fault are four prominent, narrow, more or less continuous, ENE–WSW to E–W topographic ridges, interpreted as fault-related, extending into the alluvial plain. The supporting evidence for the Khao Yai Fault is discussed below, followed by discussion of the characteristics of the narrow ridges based on a detailed Global Mapper study of the DEM images combined with a detailed study of Google Earth images, and an interpretation of the overall structural patterns.

Thailand's Department of Mineral Resources (DMR) has compiled 1:250,000 scale geological maps of Nakhon Nayok, Prachin Buri and Sa Kaeo provinces (respectively, Sinsakul and Salyapongse, 1998; Vimuktanandana and Bamroonsong, 2008; Salyapongse, 1998) and these data are in turn the basis of the 1:1 million scale geological map by the DMR (1999). None of these published geological maps shows a fault in the position of the Khao Yai Fault, or indeed any major faults, although they do show various geological contacts coinciding with it (Fig. 3b). Along the eastern part of the fault the Phra Wihan Formation crops out on both sides, but westwards the apparent displacement increases, with successively older rock units on the north juxtaposed against the Phra Wihan Formation on the south. Although the geological map (Fig. 3b) appears to indicate a substantial southward downthrow where the Phra Wihan Formation in Khao Yai National Park crops out against Permo-Triassic volcanic rocks and Permian Saraburi Limestone, it should be borne in mind that the base of the Khorat Group is a major unconformity (at this locality it is the combined Indosinian I, II and III unconformities; Fig. 2) and these older rocks subcrop that unconformable surface. The amount of vertical displacement across the Khao Yai Fault may therefore not necessarily be great.

Westwards, beyond Khao Yai National Park, the topography is less mountainous. A linear valley coincides with the mapped boundary of Saraburi Group limestone on the north and Permo-Triassic volcanic rocks on the south, and it is suggested that this apparent continuation of the Khao Yai Fault is more likely a distinct fault, on a trend which is some 25° different from the trend of the NW part of the curvilinear Khao Yai Fault (Fig. 3b). The Department of Mineral Resources (1985) shows the strike of the Saraburi Group trending parallel with the inferred fault, with steep dips (up to 80°), mostly to the SSW. Significantly, on this line the Department of Mineral Resources (1985) shows a northward-directed thrust NE of Saraburi, displacing units of the Saraburi Group at the village of Sap Bon on Highway 2. Immediately south of the thrust on the east side of Highway 2 the same Department of Mineral Resources map (1985) shows a Triassic granite outcrop some 5 km long and

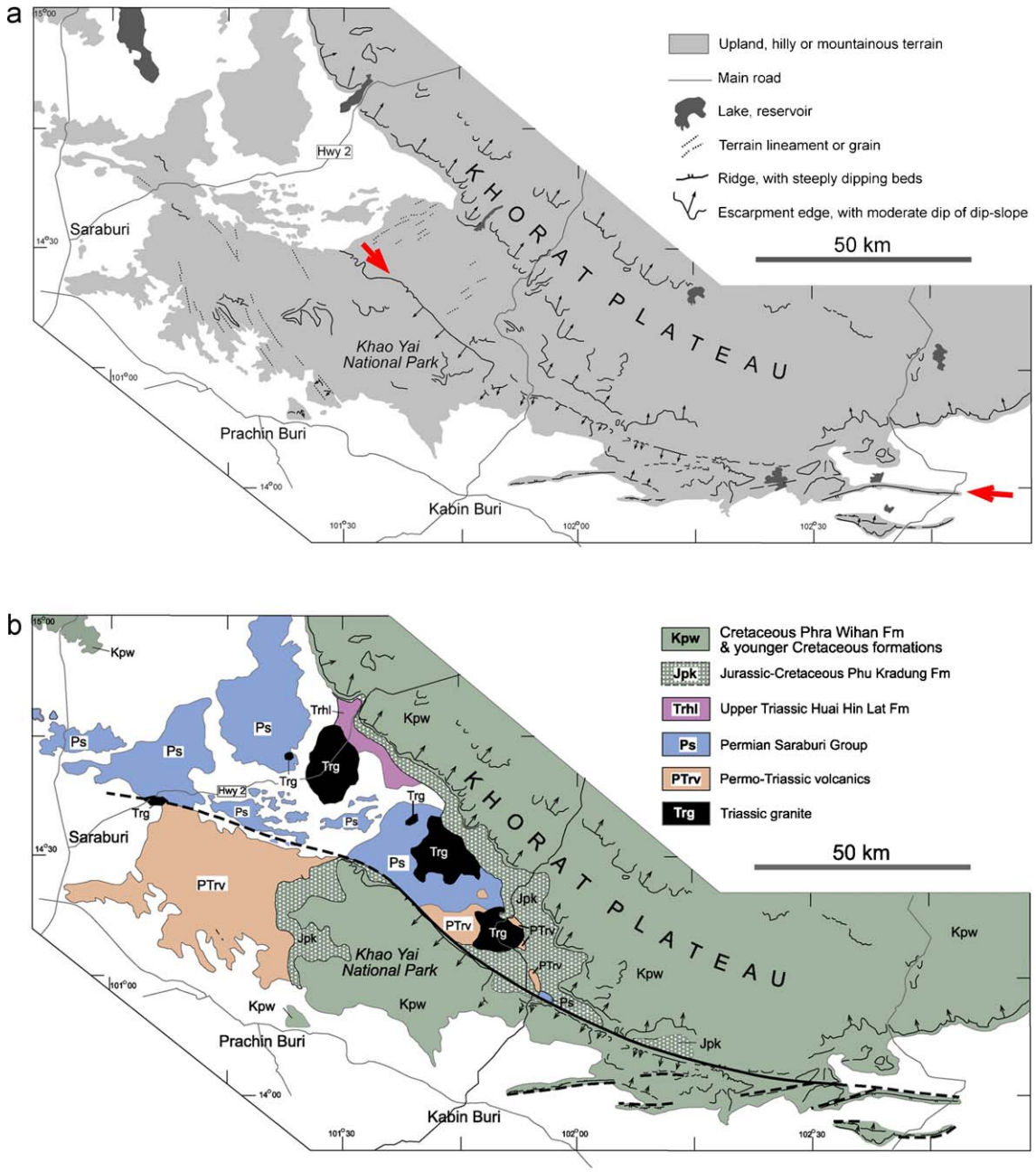


Fig. 3. (a) Photogeological map of the southern border zone of the Khorat Plateau, constructed from Google Earth data. The large arrows indicate the lineament. The narrow topographic ridges east of Kabin Buri are formed of steeply dipping beds, the double ticks indicating the direction of dip. The location is shown in Fig. 1. (b) Geological map of the southern border zone of the Khorat Plateau, based on the photogeological map shown as (a), integrated with outcrop data from the Department of Mineral Resources (1999). The thick solid black line is where the lineament is interpreted with some confidence as a fault, named here the Khao Yai Fault. The dashed lines are inferred faults, those east of Kabin Buri are inferred splay faults associated with linear ridges of steeply dipping Khorat Group beds. Further detail is shown in Fig. 5.

aligned WNW–ESE which is inferred here to be a fault sliver, an occurrence which would be consistent with the fault having a wrench component.

The largest of the narrow, steep, linear to curvi-linear ridges along the southern margin of the Khao Yai Fault are 20 and 32 km long, and about 200 m–1 km wide. On the satellite images, bedding can be seen clearly to strike parallel to the ridges. Since bedding dip direction remains consistent along the ridges it seems most reasonable to interpret the ridge-creating faults as lying on the escarpment side of the ridge, as opposed to the dip-slope side, and the faults to be either vertical or steeply dipping under the ridges (Fig. 4). The ridges display steeper dips, in the range of 40–50°, passing to the WSW, and generally decrease in dip to about 10° where the linear ridges join the main region of hilly terrane to the ENE. This suggests that fault displacement is dying out to the ENE. In the eastern part of the area the narrow, linear ridges join up with the Khao Yai Fault. Also in the eastern area there is a strong interplay between ENE–WSW and E–W to WNW–ESE trending lineations. Broad folds are developed between the intersecting trends south of the Khao Yai Fault. The ENE–WSW trending faults make an angle ranging from 16° to 25° with the Khao Yai Fault (Fig. 5).

The regional setting as well as the presence of folds and the fault pattern all suggest transpressional deformation, and the regional fault pattern (Fig. 11) suggests that the ENE–WSW faults are part of a splay geometry off the main Mae Ping Fault strand in the Kabin Buri area. The fault pattern is consistent with the ENE–WSW faults being interpreted as Riedel (R) shears within a strike-slip fault

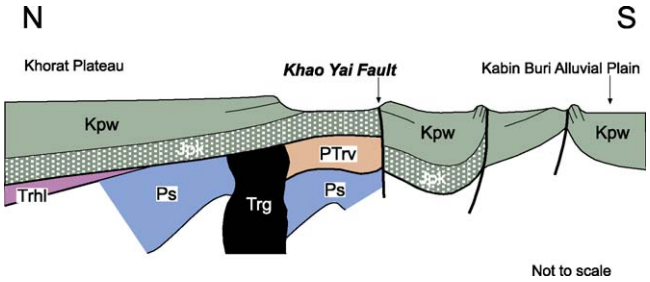


Fig. 4. Sketch section across the Khao Yai Fault and the inferred splay faults on its southern side. Line of section is approximately along longitude 102°12′ (see Fig. 3b) The pre-Khorat Group geology and the dip of the faults are conjectural.

system. However, interpreting such a fault pattern is not an easy matter since we do not know whether these faults formed at the tip of a strike-slip fault or along the main part of a strike-slip fault. It is well established that the stress orientations at the tips of strike-slip faults are different from the stress orientations along the main part of such faults (e.g. Chinnery, 1961, 1966; see review in Lovely and Pollard, 2009) (Fig. 5b and c). If the faults in the Khao Yai-Kabin Buri area are considered as having developed at a palaeo-fault tip they should not then be regarded as classic R-shears. R-shears along with R′ shears form a conjugate fault set to the maximum horizontal stress orientation, and are developed along the main parts of fault zones. The Khao Yai Fault trends about 278° in its eastern segment, swinging gradually to 300° and then to 315° still further NW in the Khao Yai National Park area. The large, broad anticline developed along the fault is best seen along the 300° trend, and is interpreted as having formed along a restraining belt caused by the change in fault trend from 278° to 300°, and is consistent with sinistral displacement. The anticline becomes broader to the NW and the Khao Yai Fault probably dies out, replaced by the inferred fault trending at 290° to Highway 2 and beyond (Figs. 3b and 11). This is consistent with displacement along the Khao Yai Fault being transferred onto the main Mae Ping Fault strand to the south via the ENE–SWS trending faults. The parallelism of the WNW–ESE trending Khao Yai Fault and the main Mae Ping Fault, linked by ENE–WSW trending faults, forms a very large strike-slip duplex geometry (Fig. 11).

2.1. History of movement on the Khao Yai Fault

Petroleum exploration over the Khorat Plateau has yielded important information on the concealed structure of the Khorat

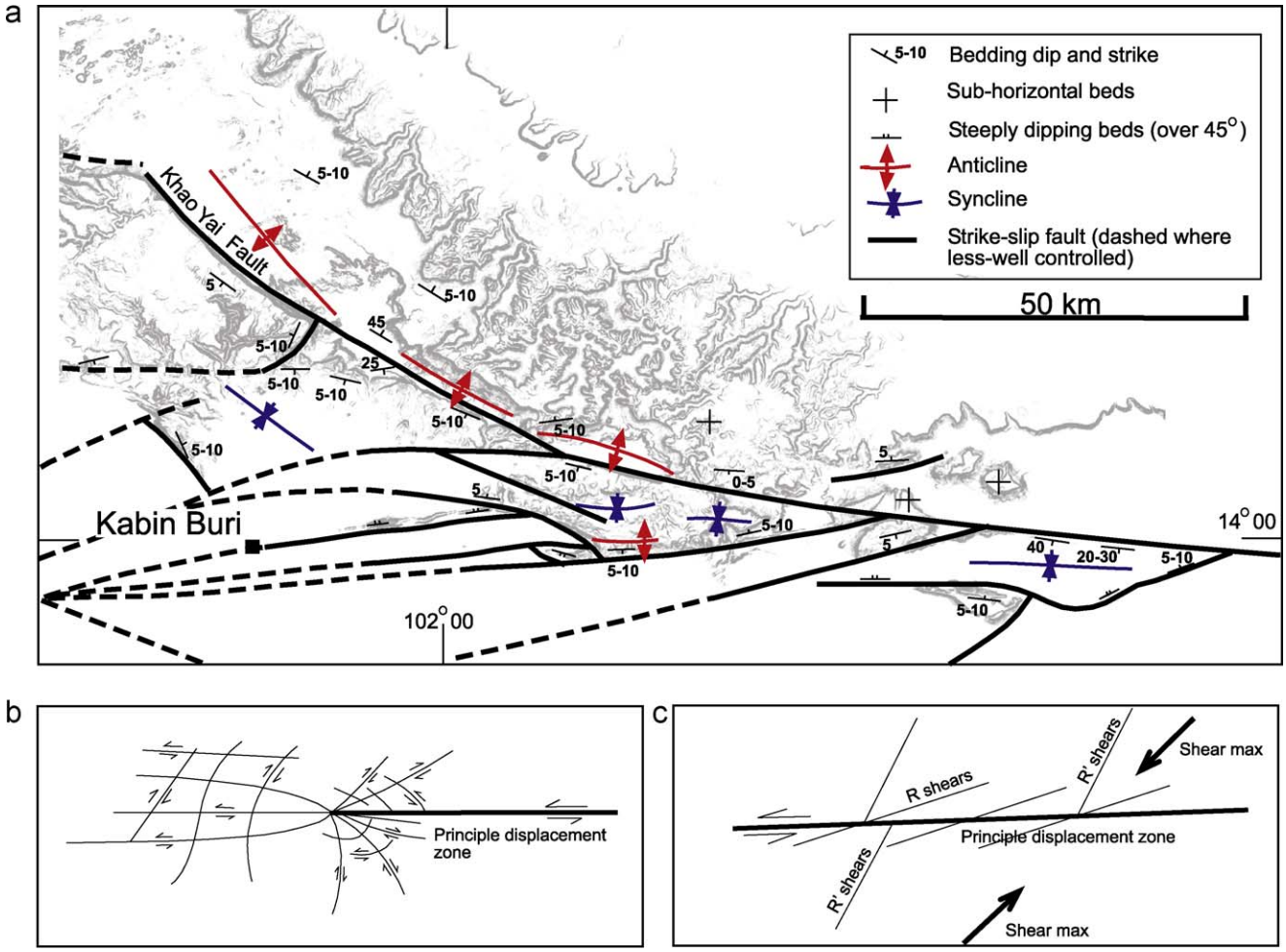


Fig. 5. (a) The Khao Yai Fault and its splays, superimposed on a DEM 'slope shader' image. The steepness of the dips has been estimated using Global Mapper and Google Earth images. The splay faults are inferred to extend beneath the alluvial cover on generally WSW–ENE trends and connect with the main strand of the Mae Ping Fault which, from other evidence, has a WNW–ESE trend and crosses the SW corner of the map. (b) Schematic illustration of the range of fault orientations that can be developed at the tip of a strike-slip fault (modified from Chinnery, 1966). (c) Schematic illustration of R and R′ shears developed along the main part of a strike-slip fault. The fault pattern in (a) can be explained by either of the models in (b) and (c), although the regionally splaying nature of the faults suggests (b) as possibly the more appropriate. A complication in the way of a simple explanation of the fault pattern is that regionally the Mae Ping Fault is known to have reversed its sense of motion from early sinistral displacement to later dextral displacement.

Group. Booth and Sattayarak (2011) illustrate the Late Triassic faults which were active at the time that non-marine, largely lacustrine, sediments of the Kuchinarai Group were being deposited, and the Khao Yai Fault invites comparison with those faults, several of which have a similar overall NW–SE trend. However, while some of those Late Triassic faults were reactivated in latest Rhaetian time and therefore displace the Lower Nam Pong Formation, Booth and Sattayarak (2011) provide no examples of such faults penetrating further up the succession. In contrast, the Khao Yai Fault cuts, however slightly, the Phu Kradung and Phra Wihan Formations. Hitherto the only documented faults on the Khorat Plateau affecting these upper units of the Khorat Group are at its northern margin, in Laos, where Lovatt Smith et al. (1996) describe a mid-Cretaceous phase of compressive folding and NW–SE trending reverse faulting beneath an unconformity at the base of the Upper Cretaceous Phon Hong Group, which is the correlative of the Maha Sarakham and Phu Thok Formations in Thailand (Fig. 2). Faulting of that age has not previously been described in Thailand although Sattayarak et al. (1991) have described seismic evidence of an angular unconformity at the base of the Maha Sarakham Formation, and folding of that age may have caused the Phu Phan Uplift resulting in the creation of two basins, to its north and south, in which evaporites of the Maha Sarakham Formation were deposited.

The pattern of pre-Khorat Group outcrops (Fig. 3b) along the line of the unnamed fault west of about 101°30′E, in particular the boundary between the Saraburi Group on the north and Permo-Triassic volcanic rocks on the south, suggests that this unnamed fault may have displaced these units before deposition of the Khorat Group; if that were also the case with the Khao Yai Fault the displacement of the Khorat Group would then have been the result of reactivation of the fault.

As will be discussed below, there was considerable Cenozoic faulting in Thailand but there is no direct outcrop evidence as to whether such later movement took place on the Khao Yai Fault. The eastern part of the Khao Yai Fault and the associated ENE–WSW fault trends clearly affect the whole of the Khorat Group up to the Phu Kradung Formation, which means the latest deformation was either Late Cretaceous or Cenozoic. As discussed in Section 1, exhumation of the Khorat Group is part of a regional deformation event, and the latest movement on the Khao Yai Fault probably occurred in that context. Upton (1999) took four apatite fission track (AFT) samples in a traverse across the area of the Khao Yai Fault and the Mae Ping Fault at Kabin Buri (Fig. 11). Those samples were part of a more regional sampling programme across the Khorat Plateau which showed that samples had experienced near total annealing during the Late Cretaceous-Early Tertiary (~70–50 Ma) and exited the partial annealing zone (~60 °C) between 49 Ma and 11 Ma. Upton (1999) concluded that the total resetting of samples THI9431 and THI9434 from the SW area of the Khorat Plateau indicated the region underwent deeper burial (or increased heat flow) compared with samples from other parts of the plateau. These samples also had the highest mean track-length values of the regional samples, which suggest a relatively fast rate of cooling. Cooling began in the Eocene based on central ages of 50 ± 2 Ma (THI9431) and 40 ± 2 Ma (THI9434). The central age close to the main Mae Ping Fault strand is somewhat younger (THI9430, 31 ± 3 Ma). Modelled results estimating the exit from the partial annealing zone are slightly earlier than the samples collected regionally (THI9431 = 43.5 Ma; THI94344 = 40 Ma; THI9432 = 30.5 Ma; Upton, 1999). These results are similar to, but somewhat younger than, the timing of exhumation for the Mesozoic Phuquoc group from Phuquoc Island in the Gulf of Thailand, determined from AFT data by Fyhn et al. (2010a), Their results determined that cooling through temperatures of 110–55 °C occurred between 60 Ma and 50 Ma.

3. Cardamomes Mountains, Cambodia

The border between Thailand's Trat Province and Cambodia runs along the crest of an escarpment of Khorat Group rocks which dip gently east. They have been mapped by Tansuwan (1998) as the Phu Kradung and Phra Wihan formations although this correlation with the type area of those units on the Khorat Plateau should be considered tentative. Where the escarpment trends NNW–SSE along the coast (i.e. south of about 12°05′N) there is some evidence that a fault may be present at the foot of the escarpment as Ridd and Wainwright (1969) noted a local steepening of the dip or a dip reversal close to outcrops of acid volcanic rock presumed to be Permo-Triassic (at 11°58′N).

This narrow outcrop belt in Thailand is the western edge of a large area of Khorat Group outcrop which forms the Cardamomes Mountain range (also called the Chuor Phnum Kravanh) in southwest Cambodia. Vysotsky et al. (1994) have reviewed the petroleum geology of Cambodia and they show this extensive area as the Kampong Saom Trough, comprising what they call the 'Upper Indosinian' or 'Upper Sandstones' of Late Jurassic to Cretaceous age, with a strong unconformity at the base of the unit. It corresponds to what Workman (1977) and earlier French workers (e.g. Gubler, 1935; Bureau de Recherches Géologiques et Minières (BRGM), 1971) called the Grès Supérieurs. The Middle Jurassic-Early Cretaceous age assignment by the French workers is consistent with the results from a borehole at the southern tip of Phuquoc Island which Fyhn et al. (2010a) report penetrated a 500 m Aptian succession.

Fig. 6 is a photogeological map constructed entirely using Google Earth terrain maps, and the outline of the Khorat Group bears a close similarity to the mapped extent of the Grès Supérieurs by the BRGM (1971) and the Upper Indosinian outcrop of Vysotsky et al. (1994). But additionally the photogeological map reveals several large structural features of the area. Folding is widespread although dips are shallow and there are large areas where the bedding is apparently sub-horizontal. At least six synclines have been mapped with widely differing trends, and there are broad anticlinal dip reversals (the axes of which are not shown because the axial areas tend to have been eroded). Numerous lineaments are evident which are inferred to be faults, mostly trending NNW–SSE but including also ENE–WSW trends.

The amount of folding and faulting appears to be greater than on the Khorat Plateau of Thailand. This might indicate that the Kampong Saom Trough experienced more (post-Lower Cretaceous) deformation than the Khorat Plateau. Or it could also be explained by the fact that the succession in the Kampong Saom Trough is thinner (1400 m according to Gubler, 1935, or 1100 m according to Vysotsky et al., 1994) than the several kilometres of Khorat Group beneath the Khorat Plateau, allowing drape over an irregular unconformity surface to be expressed in the overlying succession. Of these two possible explanations, the former is the more likely since, as will be shown below, the Kampong Saom Trough is the onshore extension of the Phuquoc-Kampot Som Basin which has undergone considerable pre-Neogene folding (albeit generally with shallow dips) and faulting.

4. Gulf of Thailand

The Kampong Saom Trough extends southward into the Gulf of Thailand where it has been described recently by Fyhn et al. (2010a) and named the Phuquoc-Kampot Som Basin (Fig. 7). Seismic survey data and exploration boreholes show that in much of this offshore area a cover of Neogene beds rests with an angular unconformity on what Fyhn et al. (2010a) informally call the Phuquoc group and which they conclude is of Late Jurassic to Early Cretaceous age, the same as onshore. In terms of facies and age the

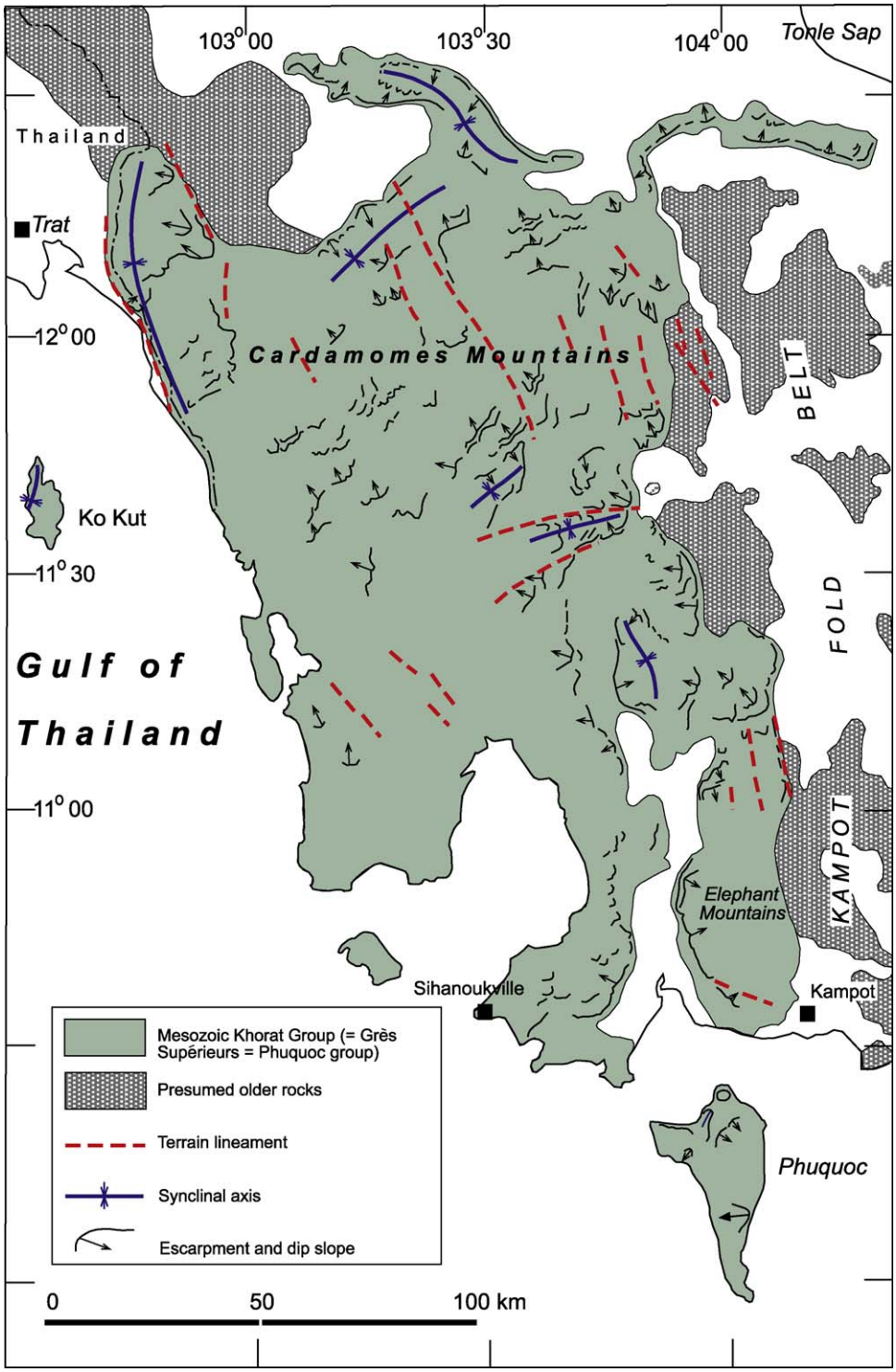


Fig. 6. Photogeological map of SW Cambodia and the islands Ko Kut and Phuquoc, constructed from Google Earth data. The narrow outcrop 'arms' lying approximately on Latitude 12°30′N have an overall WNW–ESE trend and hint at the possible presence of the Mae Ping Fault belt beneath the cover of Quaternary sediments. The Kampot Fold Belt is taken from Fyhn et al. (2010a).

Phuquoc group is clearly the correlative of the Khorat Group, and Fyhn et al. (2010a) estimate that its thickness is up to 3–4 km.

The broad structure of the Phuquoc-Kampot Som Basin is a N–S to NNW–SSE trending syncline up to about 200 km wide, cut by many generally minor faults sub-parallel to the basin axis. On its east it is bounded by the strongly faulted, N–S trending, Kampong Fold Belt which extends on land to the north and similarly bounds the Kampong Saom Trough. On its west, beneath a thickening cover of Neogene beds, it is bounded by the NNW–SSE trending Palaeogene rift basin: the Khmer Basin of Fyhn et al. (2010a) but also called the Tho Chu Graben by Fyhn et al. (2010b). Individual stratigraphic units within the Mesozoic succession show a slight regional westward thinning, but it is clear this is a structural basin, not a depositional one.

An analysis of the faults affecting the Phuquoc group mapped by Fyhn et al. (2010a) shows that almost all of them are truncated by



Fig. 7. The Phuquoc-Kampot Som Basin in the Gulf of Thailand, and its relationship with other extensive areas of Mesozoic Khorat Group and its correlatives. Small unconfirmed outcrops of Khorat Group west and southwest of Tonle Sap are shown by Vysotsky et al. (1994) and BRGM (1971) but have been omitted for simplicity. The line of question marks SW of Kampot marks the extent of seismic-reflection data, but it is considered likely that the Phuquoc-Kampot Som Basin is continuous with the Cardamomes Mountains area of Khorat Group sediments onshore Cambodia. The outlined area west of Chanthaburi shows the location of Fig. 9.

the Neogene unconformity and so are pre-Neogene; most have some degree of reverse throw on them; and the majority of the fault planes (around 80%) dip to the east. The seismic sections figured by Fyhn et al. (2010a) have the inherent character of regional seismic sections which is to exaggerate the vertical scale. Therefore, some of the faults shown on their interpreted seismic sections, while appearing to dip steeply to the east, may in fact be low-angle thrusts with dips as low as 15°, or locally lower. The faults in the eastern half of the basin form a belt which continues north, onshore, into the Kampot Fold Belt. Those in the western half

of the basin trend in a more NNW direction and align with generally east-dipping shear planes and bedding in the Triassic Noen Po Formation cropping out east of the Tha Mai Fault discussed below (Fig. 9).

The Thai island, Ko Kut (Fig. 8), falls within the larger geological province of the Kampong Saom Trough and the Phuquoc-Kampot Som Basin (Fig. 7), and it provides a window into the structure and stratigraphy. The island is composed of Khorat Group: a non-marine sandstone and mudstone succession which has yielded an important vertebrate fauna (Ridd, 1978; Cuny et al., 2007, 2010), and pollen indicating an Early Cretaceous age (Meesook, 2011). The principal structure is a syncline with an overall N–S trend occupying the north-western part of the island; dips associated with this are mostly 20° or less but at one outcrop the dip is 35°. On the southern coast of the island dips are again mostly less than about 20° although at one outcrop, thought to be close to a N–S fault, the dip is 75°. An antiformal dip reversal on the southern headland has an associated dip of c.35°. North of the island at Ko Mak, and beyond on the island Ko Chang, are extensive outcrops of Permo-Triassic acid volcanic rocks which are inferred to form part of the basement subcropping the Khorat Group. On the west coast of Ko Kut is an occurrence of Cenozoic olivine basalt (Barr and Charusiri, 2011) which has been dated by Bignell and Snelling (1977) as 8.5 ± 1 Ma, i.e. Late Miocene, and which is inferred to be part of the Neogene succession which unconformably overlies the Khorat Group.

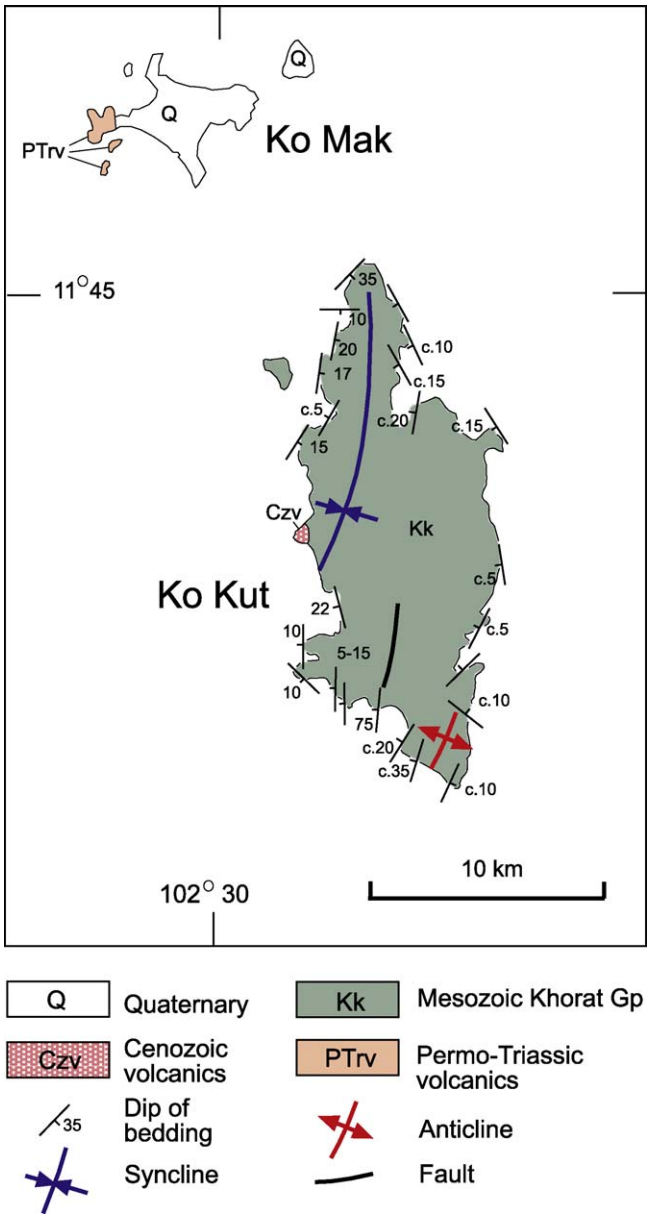


Fig. 8. Geological map of the Thai island, Ko Kut, (based on Ridd and Wainwright, 1969).

5. Khao Thalai ridge, Southeast Thailand

As described above, an escarpment of Khorat Group beds forms the boundary between Trat Province in Southeast Thailand and Cambodia, and hitherto it was thought (Ridd, 2008) that except for that escarpment the Khorat Group was absent from mainland Southeast Thailand. Another line of hills, the Khao Thalai ridge (Fig. 9), about 20 km west of Chanthaburi also consists of continental red-beds, but these were thought to be earliest Triassic or older, based on a comparison with red-beds beneath dated Triassic limestone further west (Ridd, 2008). Earlier, Bunopas (1981) had named this ridge-forming red-beds unit the Khao Thalai Red-beds and he had concluded it was Carboniferous. But it has recently been shown by Geard (2008) that these older age assignments are incorrect and that the Khao Thalai ridge is composed of Lower Jurassic or younger beds and therefore correlates with the Khorat Group. The Department of Mineral Resources had assigned it a Jurassic-Cretaceous age on its recent maps (Tansuwan, 1997; DMR, 1999) and it has now been shown to have been correct.

Geard (2008) collected a number of rock samples from Southeast Thailand and subjected them to radiometric dating at the University of Tasmania. Her sample from the Khao Thalai ridge (sample 10-4; Fig. 9) was a sandstone from the eastern slope of the ridge, north of the Rayong-Chanthaburi Highway 3. It yielded 12 zircon grains and U–Pb dating showed the youngest to be 196 ± 8 Ma, i.e. earliest Jurassic. The age of the sandstone itself is therefore Early Jurassic or younger, and the Khao Thalai Red-beds is therefore concluded to be a unit of the Khorat Group.

A radically different structural interpretation from that of Ridd (2008) is required to explain how this steeply dipping or vertical-bedded ridge of Khorat Group sandstone relates to the surrounding Triassic rocks. Its topographic expression is conspicuous, forming an overall NNW–SSE ridge from the coast at Ao Ko Nok (Fig. 9) to Latitude 12°55′N, a distance of 46 km. Its southern half is slightly more discontinuous and arcuate than the northern half, and the two segments are slightly offset at the point where the ridge is crossed by the gap followed by Highway 3.

The structure of the ridge has not been studied in detail although the southern segment, at least, is complex. The red-beds are well exposed on the foreshore at Ao Ko Nok where they strike parallel to the ridge itself, dip steeply to vertically, and cross-bedding shows they young westwards. In road cuttings nearby, where the road crosses the narrow peninsula, the rocks are again steeply dipping and here a number of high-angle faults can be seen trending NNW–SSE. In a roadside quarry about 8 km further NNW (at 12°35.64N, 101°55.25E) there is considerable deformation and one high-angle fault trends NNW–SSE, while another trends NE–SW, dips at c.70° NW, and has sub-horizontal slickensides suggesting dextral movement. These small secondary faults, which are best seen along the foreshore at Ao Ko Nok are probably rotated normal faults that formed prior to the strike-slip deformation, and do not help understand the origin of the long ridges. It is very difficult to find kinematic data for faults parallel to the ridges, due to poor exposure, but two localities display sub-horizontal slickensides with a sinistral motion.

The northern segment of the ridge, north of Highway 3, is less well exposed although it is noteworthy that Geard (2008) reports

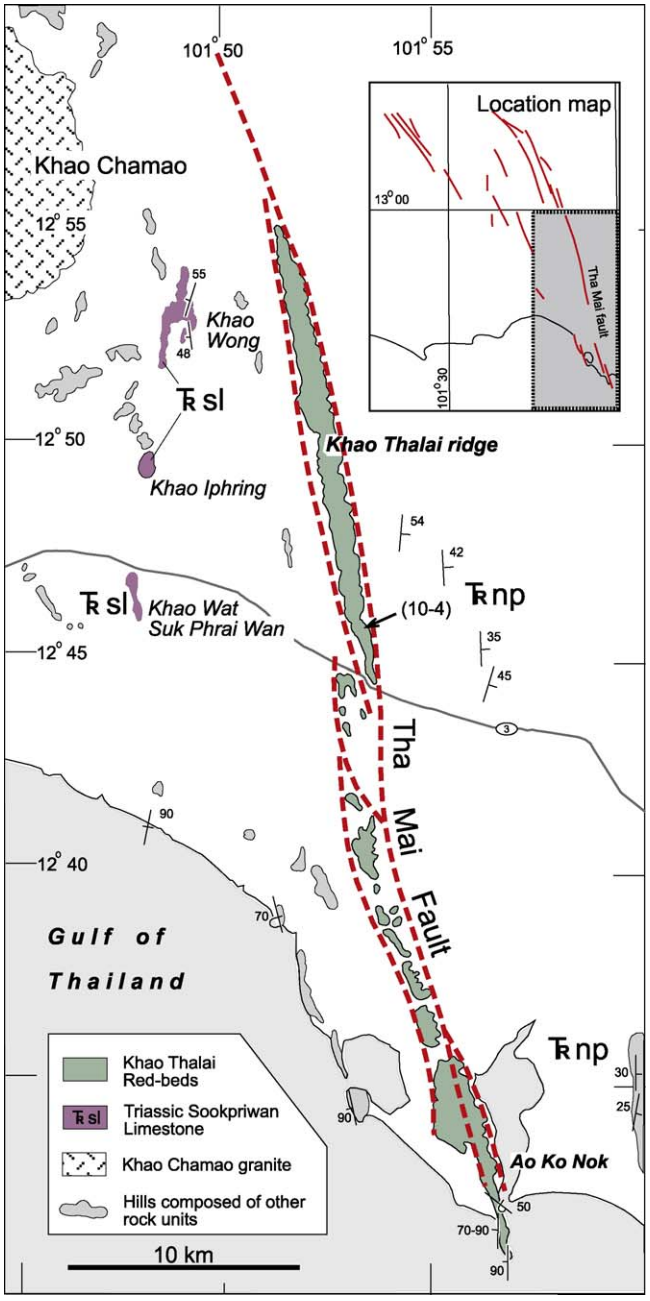


Fig. 9. Simplified geological map of the Khao Thalai ridge and adjacent areas, Southeast Thailand. The Middle to Upper Triassic Noen Po Formation (Trnp) occurs as scattered outcrops throughout the area east of the Khao Thalai ridge, while west of the ridge the rocks are more diverse and include the Triassic Sookpriwan Limestone. The Tha Mai Fault belt preserves the Khao Thalai Red-beds Formation of the upper Mesozoic Khorat Group. The dashed lines represent inferred strands of the Tha Mai Fault belt.

that the sandstone at her sample 10-4 locality (Fig. 9) youngs to the east, the opposite of the younging direction at Ao Ko Nok. Recent ditch excavation on the east side of the ridge, 1 km north of Highway 3, revealed a highly shattered, brecciated, and mineralised fault zone, supporting the interpretation of a fault zone bounding the east side of the ridge. No kinematic indicators were present.

East of the Khao Thalai ridge the terrain is generally subdued but one of us (MFR) has observed dark grey shale of the Noen Po Formation cropping out in the stream to the east of sample 10-4, and quarries in a N–S line of hills further south (east of Ao Ko Nok) provide good outcrops of similar dark grey shales as well as minor black pyritic shale and minor graded sandstones. A noticeable feature of all of these outcrops east of the ridge is that they show more or less intense shearing, the dip of the shear planes and of the bedding being consistently to the east at between about 25° and 55°. Hada et al. (1999) have described Middle to Late Triassic radiolarian faunas from other locations in this tract of country.

West of the Khao Thalai Ridge the rocks are more diverse. On the coast 10 km NW of the Ao Ko Nok headland is a similar prominent headland formed of steeply dipping or vertical cross-bedded sandstone, apparently younging to the west. This sandstone has a high volcani-clastic content suggesting it is not equivalent to the Khao Thalai Red-beds. It probably correlates with an older succession seen by Fontaine and Vachard (1981) and Ridd (2008) underlying the Sookpriwan Formation limestone which forms hills along the regional strike to the NNW. This massive grey limestone resembling the Saraburi Group has yielded foraminifera with a possibly brackish-water character which Fontaine and Vachard (1981) determined as being of Scythian-Anisian age; it was named the Sookpriwan Formation by Chaodumrong (1992) and the Khao Wong Formation by Tansuwan (1997). The Sookpriwan Formation is exposed in a large quarry at Khao Wat Suk Phrai Wan (also called Khlong Din). On the eastern side of the quarry the strike swings from 260° and dipping 45°S, to a strike direction of 180° with a vertical dip.

A 1 m wide fault zone filled with phyllonites is present on the eastern margin; there the foliation is vertical and strikes at 176° and sinistral striations plunging 5° 180° are present. This outcrop is the clearest indication found that the steep dip of the bedding sub-parallel to the ridges is related to sinistral strike-slip faults.

It is clear that the upper Mesozoic Khorat Group which forms the Khao Thalai ridge is flanked by suites of Triassic rocks: deep marine shales and probable turbidites of the Noen Po Formation on the east and shallow marine or even brackish-water fossiliferous limestone, the Sookpriwan Formation, overlying red-beds on the west. Ridd (2008) inferred a major fault, the Tha Mai Fault, separating the Noen Po Formation from the continental and shallow-marine rocks, and although the age of the Khao Thalai Red-beds has been revised the need to infer such a fault remains and, if anything, is strengthened. However, it is now proposed that the Tha Mai Fault is actually a fault belt in which a sliver of Khorat Group, apparently down-faulted, has been preserved. Abundant faulting is observed in the southern segment of the Khao Thalai ridge, although the faults shown in Fig. 9 are schematic.

There is no direct evidence of the age or sense of movement on the Tha Mai Fault. It was clearly Late Cretaceous or younger, and its long and remarkably linear character suggest that it is a wrench fault, probably one of several splays of the Three Pagodas Fault. Originally recognized in Western Thailand (Ridd, 1971) it has since been shown that the Three Pagodas Fault belt comprises a number of strands (Morley, 2002, 2004; Morley et al., 2011). Ridd (2009) has provided evidence that one strand connects with the Khlong Marui Fault in Peninsular Thailand and has suggested that it marks the reactivated eastern boundary of a Carboniferous-Early Permian rift which formed as the Sibumasu Block separated from Gondwana. The Khlong Marui Fault and the related Ranong Fault in Peninsular Thailand have been shown by Watkinson et al. (2008) to have had a complex history of dextral shear (a pre-Campanian phase followed by a Late Campanian-Palaeocene phase) succeeded by a phase of sinistral shear (post Early Eocene) and finally another phase of dextral shear (at about the Oligocene-Miocene boundary). Based on a prominent aeromagnetic lineament Milsom (2011), Morley (2002) and Charusiri et al. (2002) have proposed that another branch of the Three Pagodas Fault runs approximately through Bangkok before swinging SSE through Southeast Thailand. The structural grain of Southeast Thailand is predominantly NNW–SSE (Fig. 10) and various

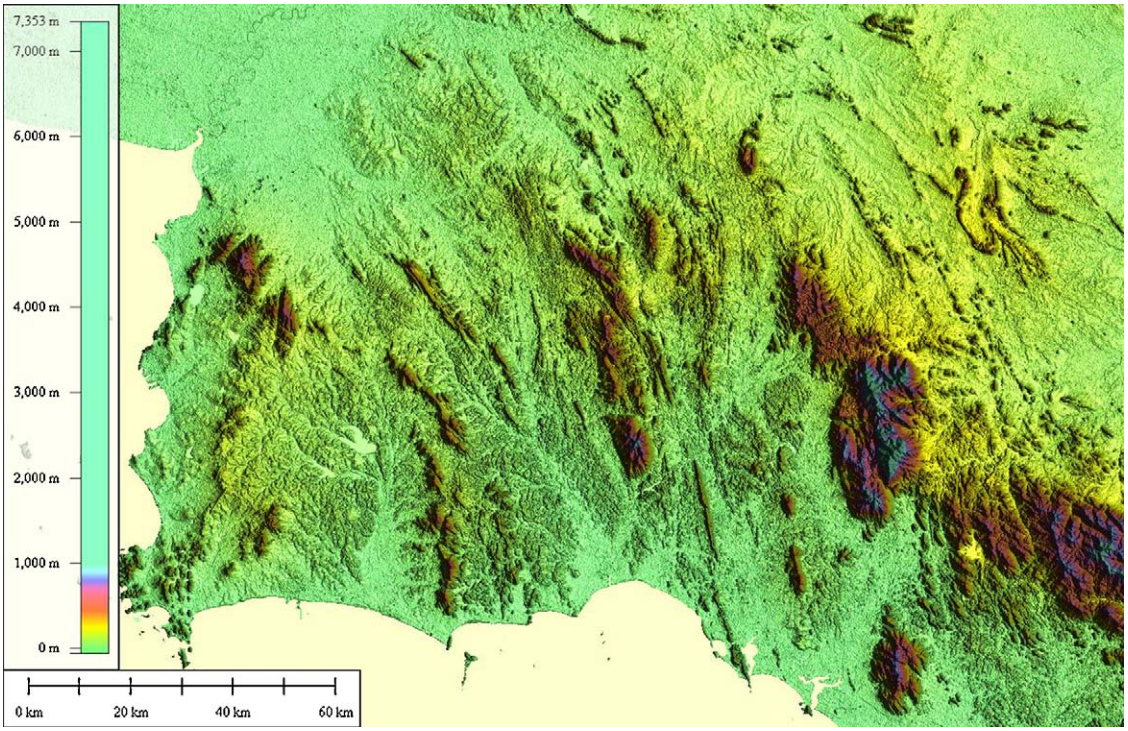


Fig. 10. Digitised elevation model (DEM) of Southeast Thailand showing the structural grain and prominent lineaments which are interpreted as strike-slip faults (see Fig. 11). Place names and other information have been omitted to allow an unobstructed appreciation of the structure.

lineaments have been suggested for the continuation of the Three Pagodas Fault, but most commonly one strand is shown running close to the town of Klaeng in Rayong Province, where it is referred to as the Klaeng Fault (e.g. Sone and Metcalfe, 2008; Ueno and Charoentitirat, 2011). Morley et al. (2011) cite evidence that substantial sinistral wrench movement took place on the Three Pagodas Fault in the Eocene and that a lesser phase of dextral wrench movement occurred in the Oligocene-Early Miocene. The strongest evidence for sinistral motion is the prominent, isolated, NW–SE trending ridge at Nong Yai, 55 km NW of Klaeng (Fig. 11). There the exhumed lower crustal part of the Three Pagodas shear zone is a ridge of steeply dipping, NW–SE trending, sinistrally deformed mylonites within Indosinian migmatitic gneisses (Lacassin et al., 1997; Geard, 2008). Ur–Pb laser-ablation-inductively-coupled-plasma-mass-spectrometry dating of the mylonites by Geard (2008) yielded a monazite age of 42.54 ± 0.88 Ma, and two syn-kinematic sphene ages of 35.5 ± 3.1 Ma and 37.8 ± 4.8 Ma. There are many linear topographic features present on satellite and DEM images (Fig. 10), which can be interpreted as part of a complex splaying array of faults related to the Three Pagodas Fault (Fig. 11). Some of these strands cut Palaeozoic rocks, but others cut Khorat Group and Eocene granite, and so indicate the faulting was Eocene or younger. The Tha Mai Fault forms the easternmost splay related to the Three Pagodas Fault.

5.1. Juxtaposition of facies belts by the Tha Mai Fault

As mentioned above, the Tha Mai Fault coincides with the boundary between continental and shallow-marine Triassic rocks on the west and the deep-marine, Triassic, Noen Po Formation on the east. The fauna of the limestone has been described by Fontaine and Vachard (1981) who, as well as assigning it a Scythian to Anisian age, mention that it may have had a brackish-water depositional environment. They mention also that it overlies a red-beds unit which also suggests a less than fully marine environment. Ridd and Wainwright (1969) had earlier described a red-beds succession beneath the limestone. Fontaine and Salyapongse (1999) plotted the distribution of this Sookpriwan Limestone and its correlatives in a belt about 20 km wide, extending from Khao Suk Phrai Wan (Sookpriwan) beside Highway 3 (Fig. 9) in the south, to the northernmost outcrop found, at Khao Tham Rat (13°20′N, 101°40′E).

Within that same belt of country, in the vicinity of Khao Iphring (Fig. 9), a limestone unit possibly correlating with the Triassic Sookpriwan Limestone is underlain by shale and limestone which have yielded the shallow-marine Permian brachiopod *Leptodus* sp.[2] (Salyapongse et al., 1997; Fontaine and Salyapongse, 1998); (nearby, as mentioned above, the same limestone is underlain by red-beds). In their recent comprehensive review of Thailand's Carboniferous and Permian rocks Ueno and Charoentitirat (2011) comment on the exoticism of these Permian rocks, suggesting that they more closely resemble the Ngao Group of Northern Thailand than any unit closer to hand. Further evidence that this belt of scattered shallow-marine rocks is of limited extent comes from an occurrence of thin-bedded shale, chert and sandstone only 20 km NW of Khao Iphring, at Wang Chik, where a latest Permian radiolarian fauna has been described by Sashida et al. (2000). Although Sashida et al. (2000) describe the facies as pelagic, K. Ueno and Y. Kamata (personal communication) note the presence of sandstone beds and clastic grains within the chert and suggest it is more likely of hemipelagic origin. Clearly it is of deeper-water origin than the *Leptodus*-bearing shale at Khao Iphring.

An unusual and localised group of outcrops of Carboniferous rocks occurs at the northern end of this same belt of shallow-marine Triassic limestone occurrences (Fig. 11). From within a 10 km radius centred on a point at about 13°20′N 101°40′E, Bunopas (1981) described a succession of shale, sandstone, chert and limestone with a shallow-marine fauna of possible Carboniferous age. Further work by Fontaine and Salyapongse (1997a,b)

[2] Although identified originally as *Leptodus* sp. and cited by subsequent authors as that genus, Dr. Masa Sone (pers. comm.) considers this brachiopod to be another lyttoniid genus, *Oldhamina*, typical of the Late Permian.

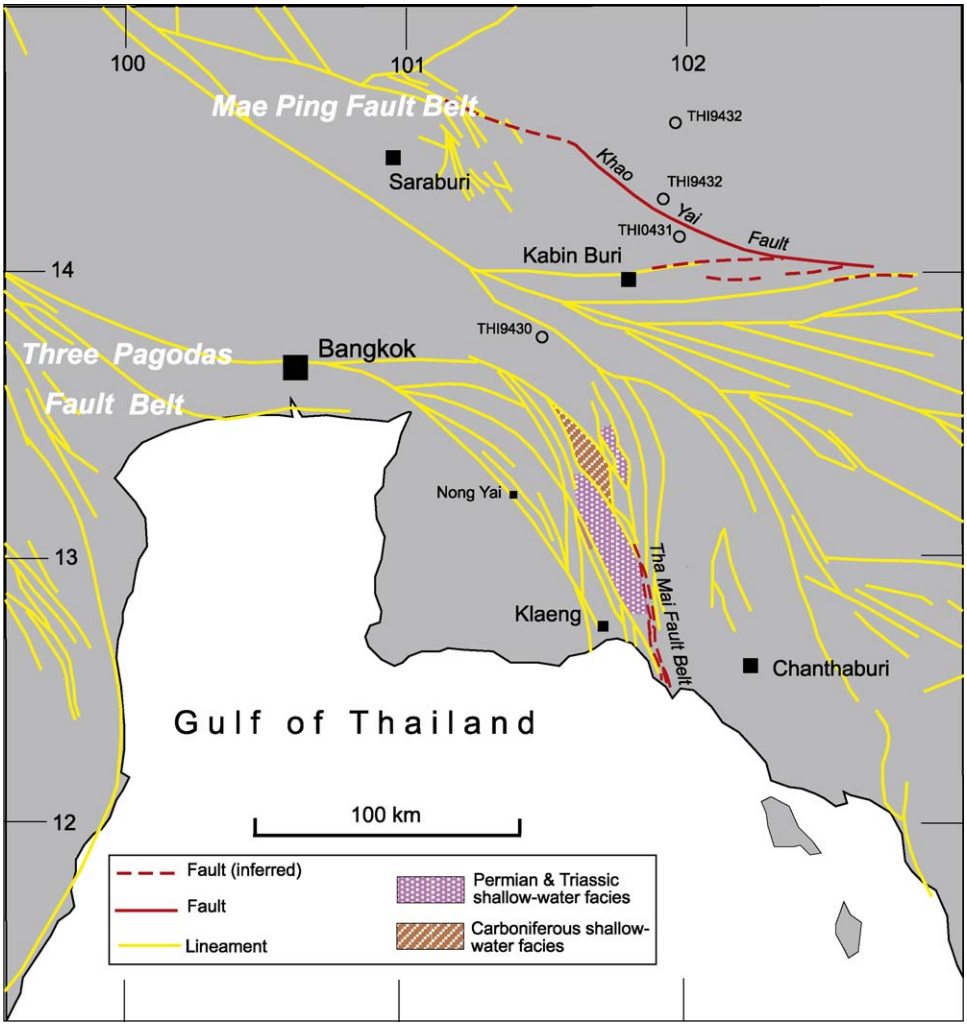


Fig. 11. Lineaments of Southeast Thailand, showing the Tha Mai and Khao Yai Faults and how their presence supports the inference that the lineaments are themselves faults. The highlighted areas west of the Tha Mai Fault represent the blocks on which shallow-marine rocks of Triassic, Permian and Carboniferous age have been described. Sample numbers (THI9430, etc.) are apatite fission-track samples of Upton (1999).

and Fontaine et al. (1999) provided a fuller picture of the unusual fauna of foraminifera, corals, calcareous algae, bryozoa and brachiopods and confirmed their age as Late Visean to Bashkirian. Recently, Ueno and Charoentitirat (2011) have commented on the uniqueness of the succession and its fauna and suggest its closest correlatives are in carbonates occurring on the southwest and west side of the Indochina Block.

Thus there is evidence from the Triassic, Permian and Carboniferous that within Southeast Thailand there is a localised belt of shallow-water rocks, and that the eastern boundary of that belt is the Tha Mai Fault. The deep-water facies of the Triassic Noen Po Formation east of the fault is closely associated with the Sa Kaeo-Chanthaburi Accretionary Complex of Chutakositkanon and Hisada (2008), and it is interpreted by Ueno and Charoentitirat (2011) as having been deposited in the southern extension of the Nan back-arc basin in its closing stages. The Tha Mai Fault is therefore seen as a major feature bounding the western side of that tectono-sedimentary province. Arguably it meets the definition requirements of a terrane boundary, although Ueno and Charoentitirat (2011) stop short of that and, recognizing its distinctiveness, refer to the region west of the Tha Mai Fault (and bounded on its west by a NNW–SSE fault in the vicinity of Klaeng) as the Klaeng Zone.

Fig. 11 integrates the Tha Mai Fault into a map of the lineaments in Southeast Thailand prepared by one of us (CKM), and the fault's presence gives strong support to the supposition that the other lineaments are also faults. Note too how the Khao Yai Fault fits with the broad pattern of lineaments. The Tha Mai Fault is apparently the easternmost of the various strands of the Three Pagodas Fault belt. Adjacent to the Tha Mai Fault on its west are shown the lineament-bounded segments (inferred to be fault blocks) on which occur the shallow-marine Triassic, Permian and Carboniferous beds described above. It is suggested that they are allochthonous and that they were emplaced by wrench movements on the Tha Mai Fault, a strand of the major regional feature the Three Pagodas Fault.

6. Conclusions

1. The Khao Yai Fault is of a type which is rare on the Khorat Plateau. While its approximately E–W trend is typical of the subsurface faults which delineated Permian and Triassic basins and facies belts, this fault is unusual insofar as the Upper Jurassic-Cretaceous units (Phu Kradung and Phra Wihan Formations) of the Khorat Group were affected, perhaps through re-activation. In this respect the Khao Yai Fault is similar to the mid-Cretaceous faults in Laos described by Lovatt Smith et al. (1996). However, whereas movement on those Laotian faults is

thought to have ended in the Late Cretaceous there is no evidence that the Khao Yai Fault became inactive then.
2. It is suggested here that the Khao Yai Fault and its associated splay faults are strands in a strike-slip duplex of the much bigger Mae Ping Fault belt which has been postulated (largely on geophysical evidence) to extend south of the Khorat Plateau into Cambodia.
3. In the northern Cardamomes Mountains of Cambodia, folds on WNW–ESE trends give some support to the presence of a larger-scale dislocation in the area, probably the further eastward extension of the Mae Ping Fault belt.
4. The offshore Phuquoc-Kampot Som Basin has a structural style that is different from the onshore Cardamomes Mountains and Khorat Plateau. It is folded and strongly faulted on NNW–SSE trends. The predominantly eastward dip of the fault planes may reflect the fabric of the basement, since onshore along the regional strike (in Southeast Thailand) an eastward dip of bedding and shear planes is a conspicuous feature of the Triassic Noen Po Formation.
5. Radiometric dating (Geard, 2008) has shown that the Khao Thalai Red-beds are a unit of the Khorat Group. It is suggested that the 46 km long, linear, outcrop of these rocks is a downfaulted sliver within the Tha Mai Fault belt and that, furthermore, this fault is a splay of the Three Pagodas Fault.
6. The faults, folding and exhumation described here appear to be a very significant part of Palaeogene transpressional deformation that affected a large part of the Gulf of Thailand and surrounding landmasses. Deformation began around 50–60 Ma and continued in some places to around 30 Ma.
7. Cenozoic deformation commonly appears to have reactivated older trends. Further work in Southeast Thailand is recommended to investigate the Tha Mai Fault belt, its detailed structure, and whether it played a part in emplacing the apparently exotic shallow-marine Carboniferous, Permian and Triassic rocks on its west side, as suggested here.

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